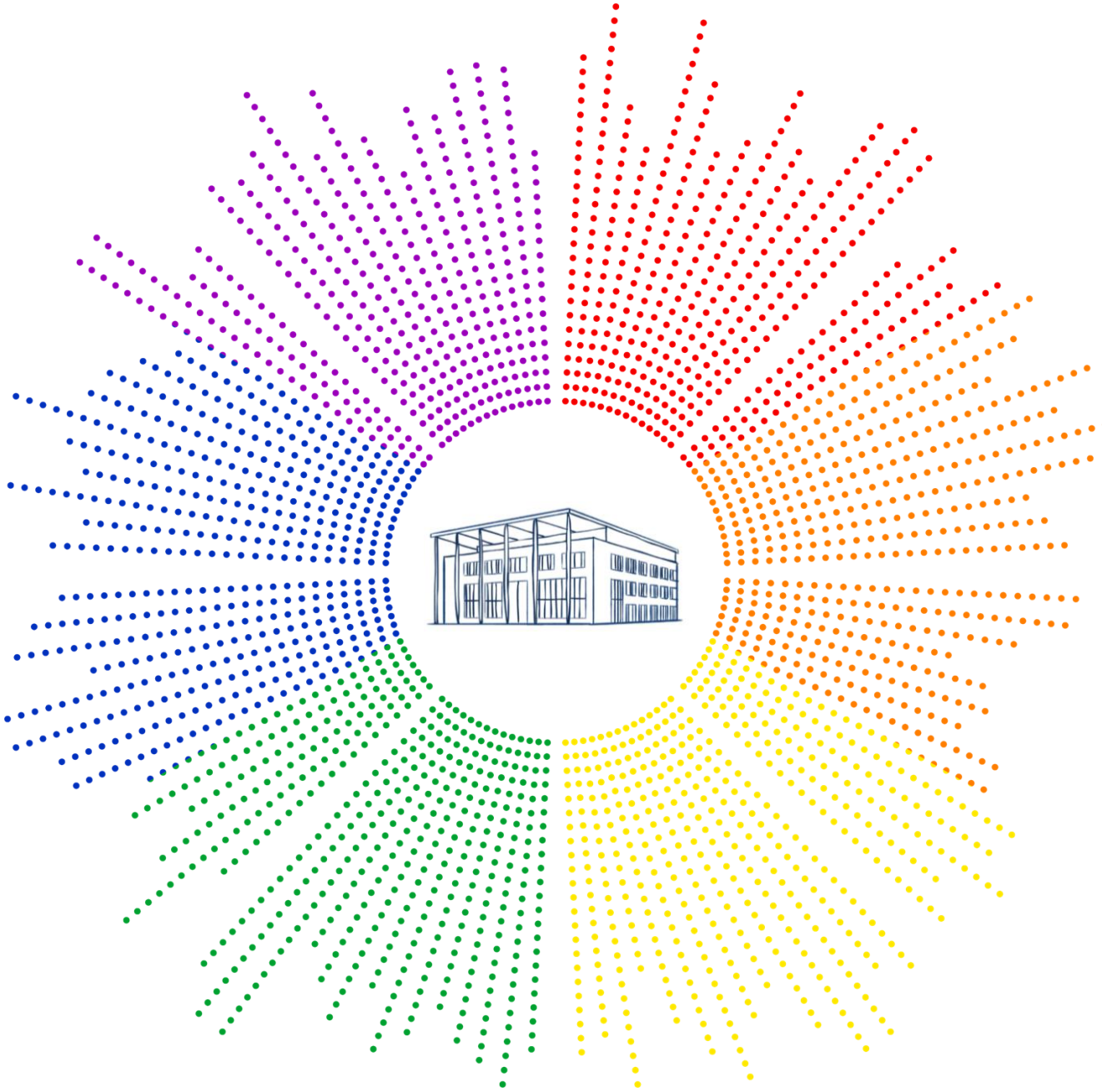
Spatial AI Lab:
Interpretation of 3D Autoregressive Transformers

Ethan Cao

10/02/2026

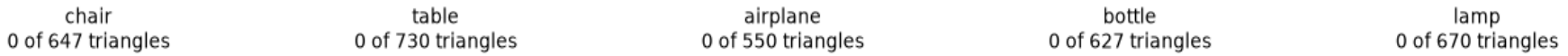


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Research Proposal

What's inside the black box?

- **Context:** Substantial research have been made into the development of **autoregressive** 3D generative models, i.e. MeshGPT and followed by a series of variant and improved models.



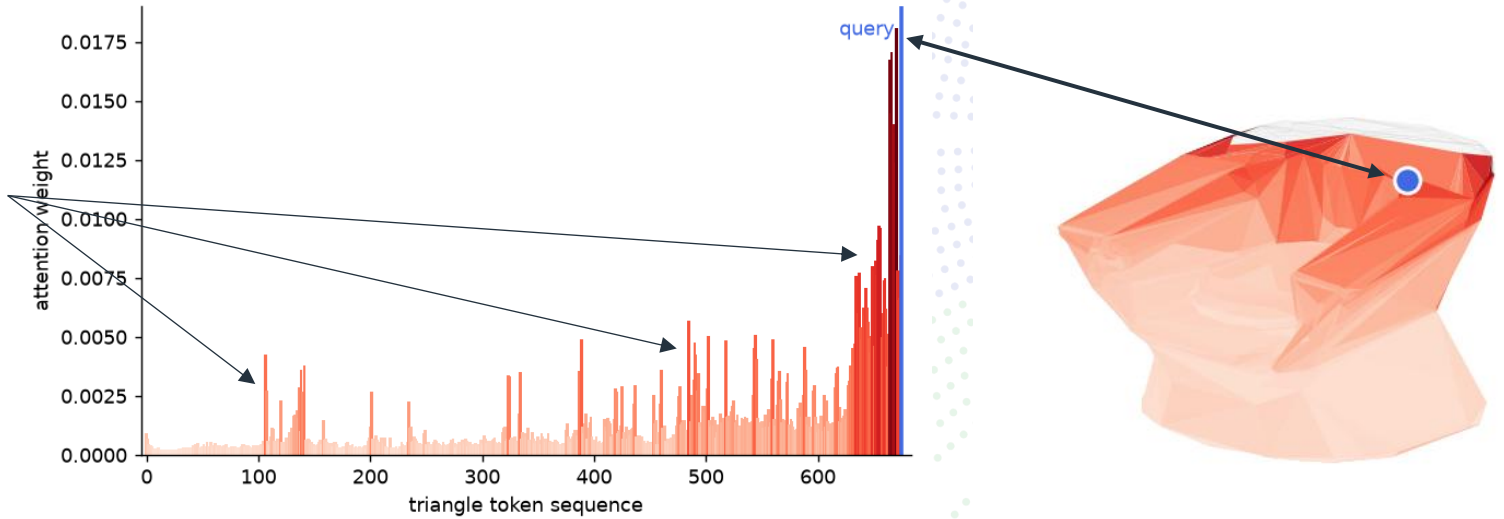
- **GIF:** Demonstration of autoregressively generating triangular meshes

Research Proposal

How to interpret 3D Autoregressive Mesh Transformers?

- **The Gap:**
  - There is no published study of what individual attention head do in mesh generation transformers. Recent models are developed with assumptions of attention mechanisms rather than systematically and quantitatively measuring it.
- **Opportunity:**
  - In language models, measuring attention head behavior led directly to a design of faster decoding and memory caching features. DuoAttention (ICLR 2025).
  - The same approach and inspiration can be applied to enhance autoregressive mesh transformers.

We examine the question of what query triangles **attend to** in the early token sequence, and **why?**

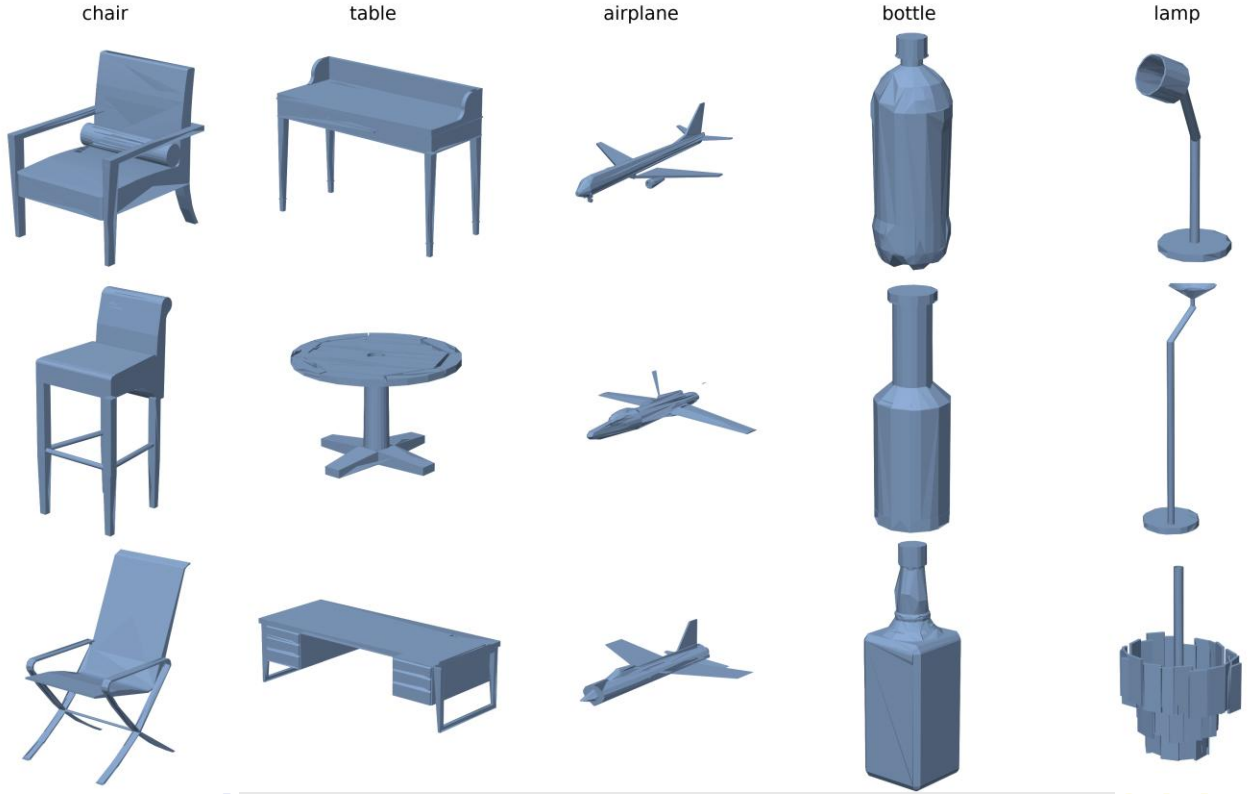


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Experimental Method

Data Set & Model of Choice

- **Choose 500 ShapeNet shapes from 5 shape categories:** chair, table, airplane, bottle, lamp
- **Choose 4 model families** to experiment with:
  - **MeshGPT:** learned-codebook decoder only transformer
  - **MeshXL:** raw coordinate prediction decoder only transformer
  - **DeepMesh:** "Point-cloud to mesh" generators with hourglass transformer structure
  - **MeshMosaic:** "Part by part" mesh generators, generates in small patches and fuses together into one mesh object



- 5 categories of shape examined

Experimental Method

Feed: run inference by passing 500 ShapeNet meshes into each model in one forward pass

Record: at one query triangle, save the attention weight each head gives to every earlier triangle

Select: select attended triangles that holds significant attention weight assigned by a head.

Group: group ShapeNet triangles into clusters by different their triangle properties

Compare: for each head, compare what category (cluster) of triangles do the attended triangles fall into.

- Potentially find generalizable patterns across the 4 models

Model Enhancements:

Hard code attention head with rules/policies

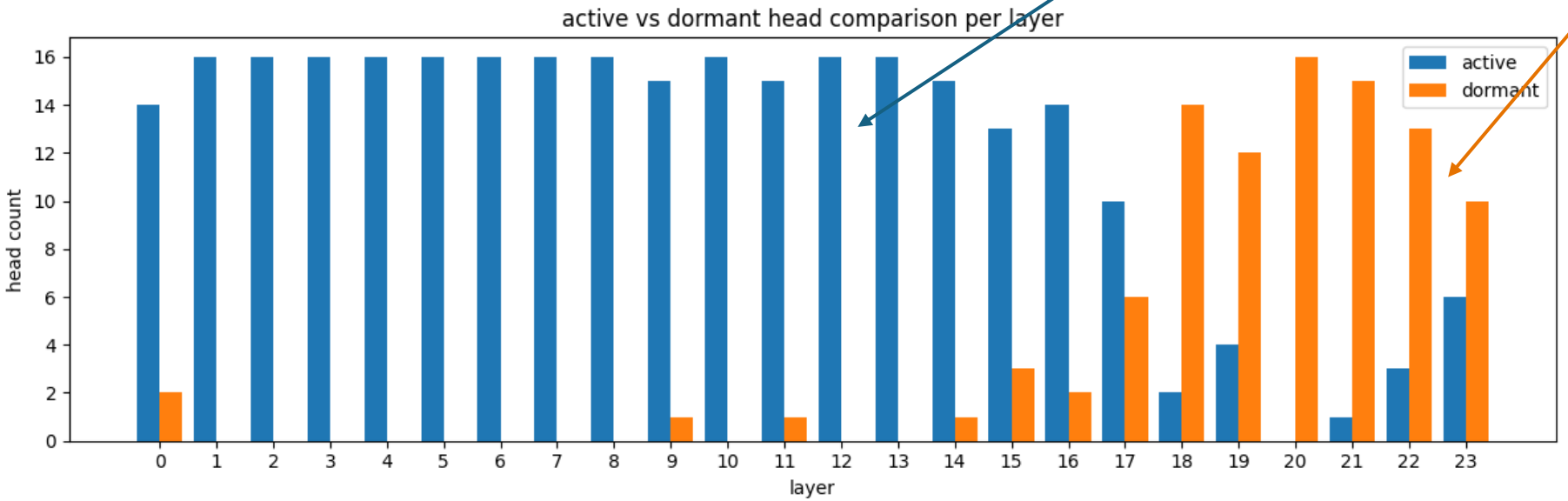
Etc…

Experimental Method

Examining “active” attention heads

- **Active-Dormant head classification:**
  - **Dormant** heads attend largely to the “beginning of sequence” token or to the very first few tokens of the token sequence.
  - **Active** heads pay attention to nearby tokens, or any tokens that are not the BOS or initial tokens.

MeshGPT: most attention heads become ‘active’ at earlier layers and become ‘dormant’ at later layers.

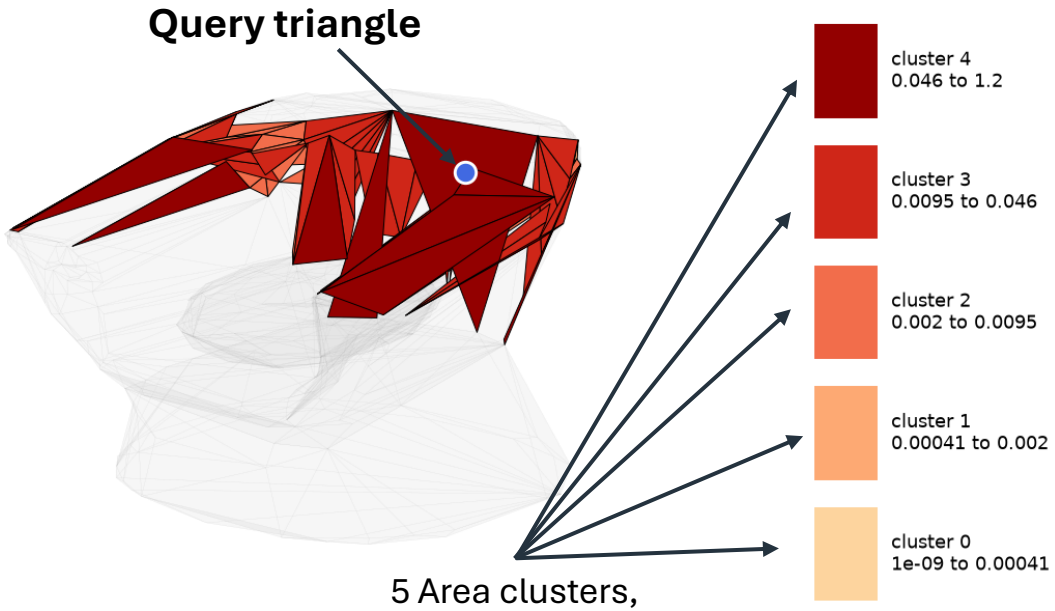


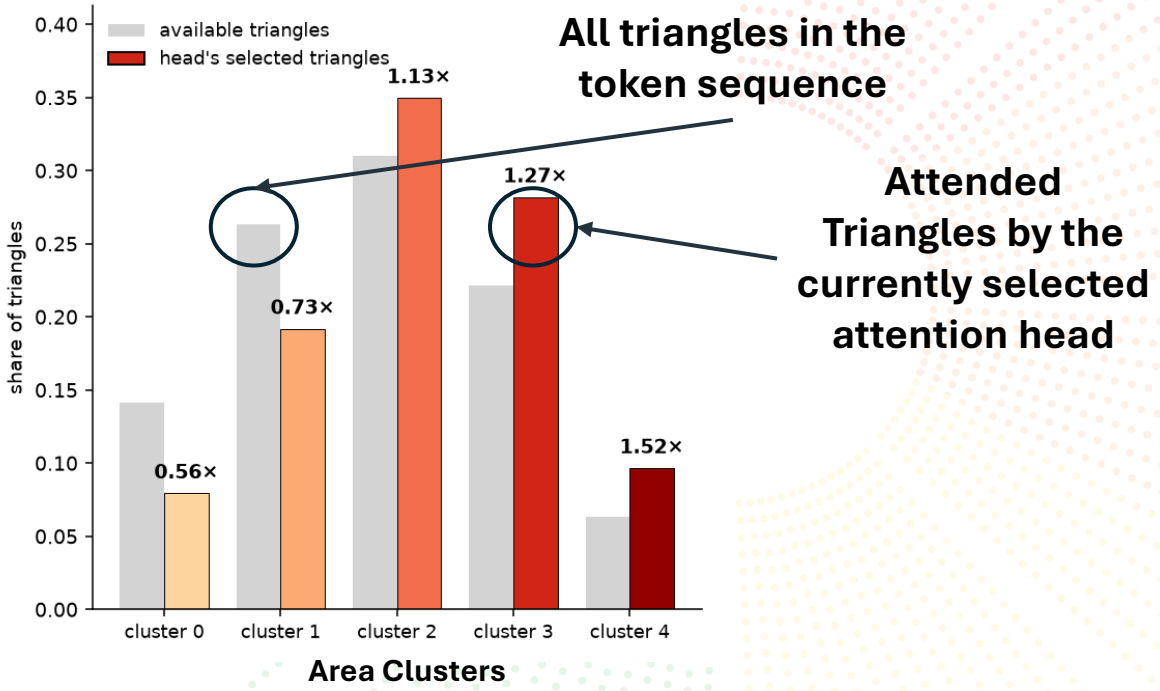
- **We examine ‘active’ heads as they present more informative insights by attending to a variety of tokens / triangles**

Experimental Method

What 'kind' of triangles do the heads attend to?

- **Selection of triangles for examination:**
  - Cluster the 500 chosen shapes by their triangular mesh features: **Aspect Ratio, Area, X-axis, Y-axis, and Z-axis normal vector orientation** into 5 clusters for each property.
  - Compare which of clusters do the attended triangles fall into.





Current Findings (WIP)

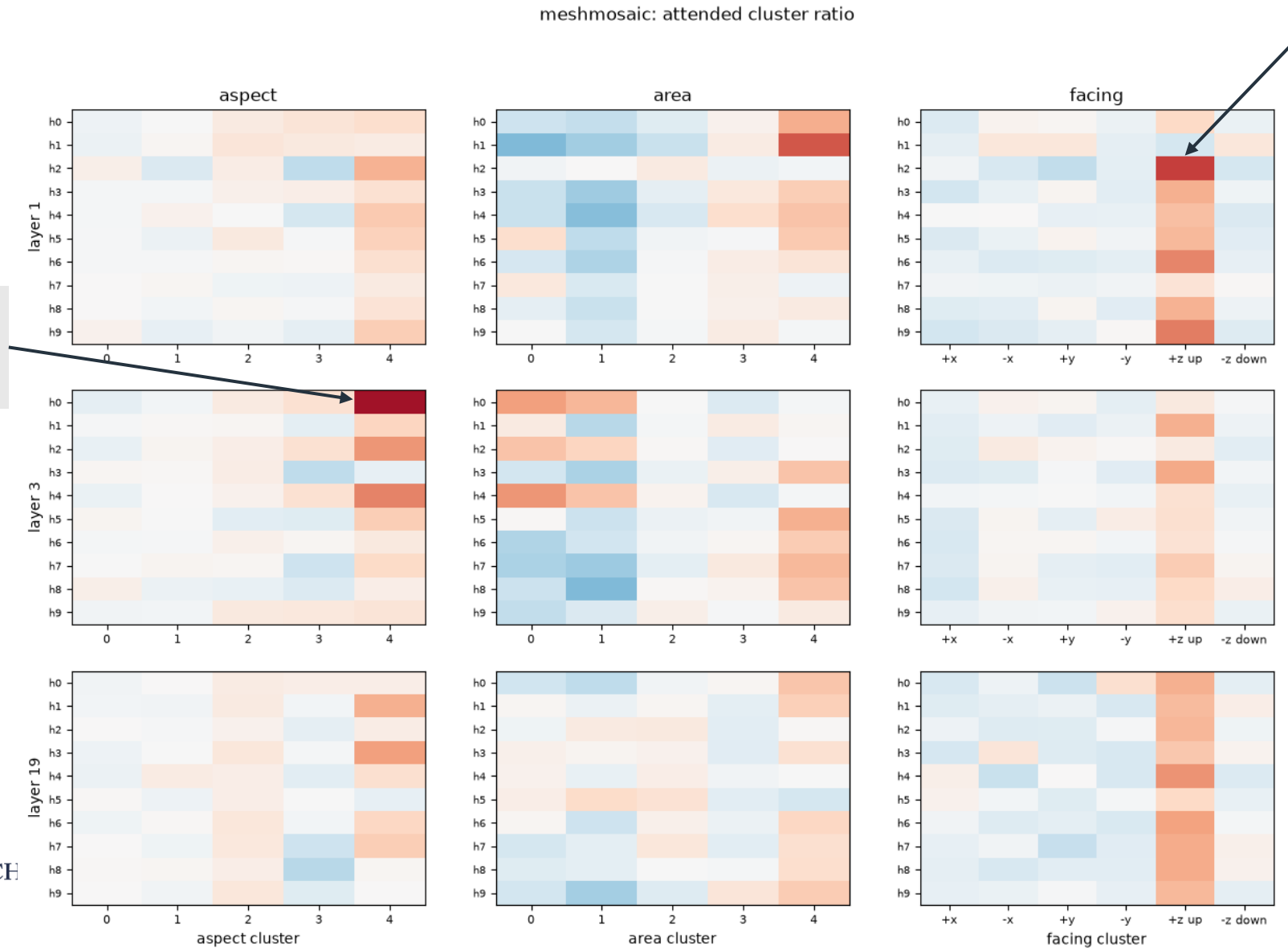
TLDR

- Across the four models, individual attention heads show **tendencies to prefer specific shapes** of triangles. (differing by the model, head, and transformer layer)
- I.e. certain heads strongly favor **large area triangles** and some other favor **thin and elongated** shaped triangles.
- These shape preferences have been shown to be **consistent at different query positions** across the entire generated mesh token sequence.
- These behaviors are **inconsistent across different transformer layers** (we classify into early, middle, and late layers)

MeshMosaic Findings (WIP)

*Head 2 strongly favors Triangles **facing the z-axis orientation** at Layer 1*

*Head 0 strongly favors **Thin & Elongated** Triangles at Layer 3*



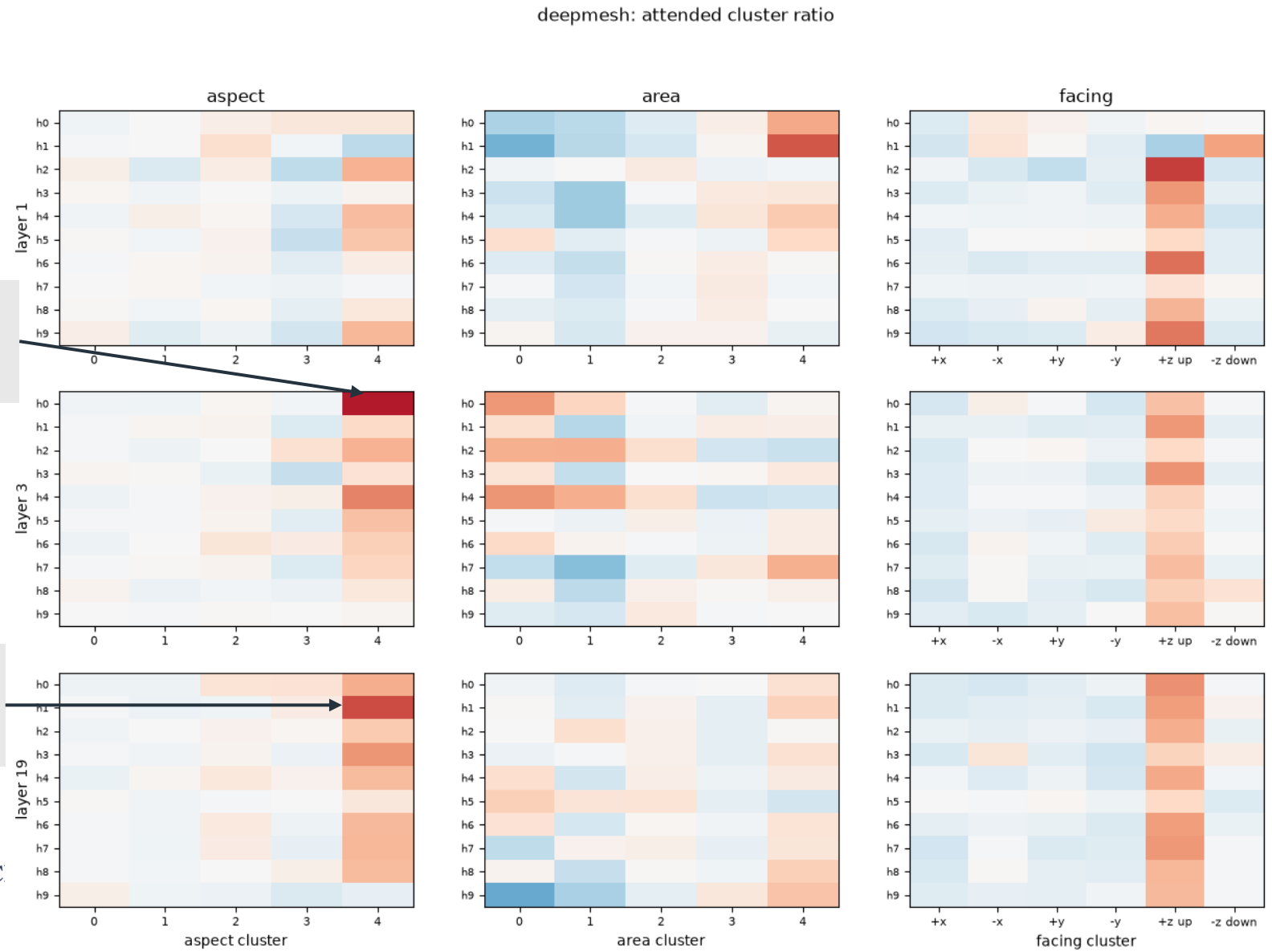
Attention Head Preferences

>1 means favor

1< means avoid

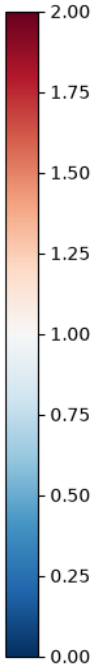
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DeepMesh Findings (WIP)



*Head 0 strongly favors **Thin & Elongated** Triangles at Layer 3*

*Head 1 strongly favors **Thin & Elongated** Triangles at Layer 19*



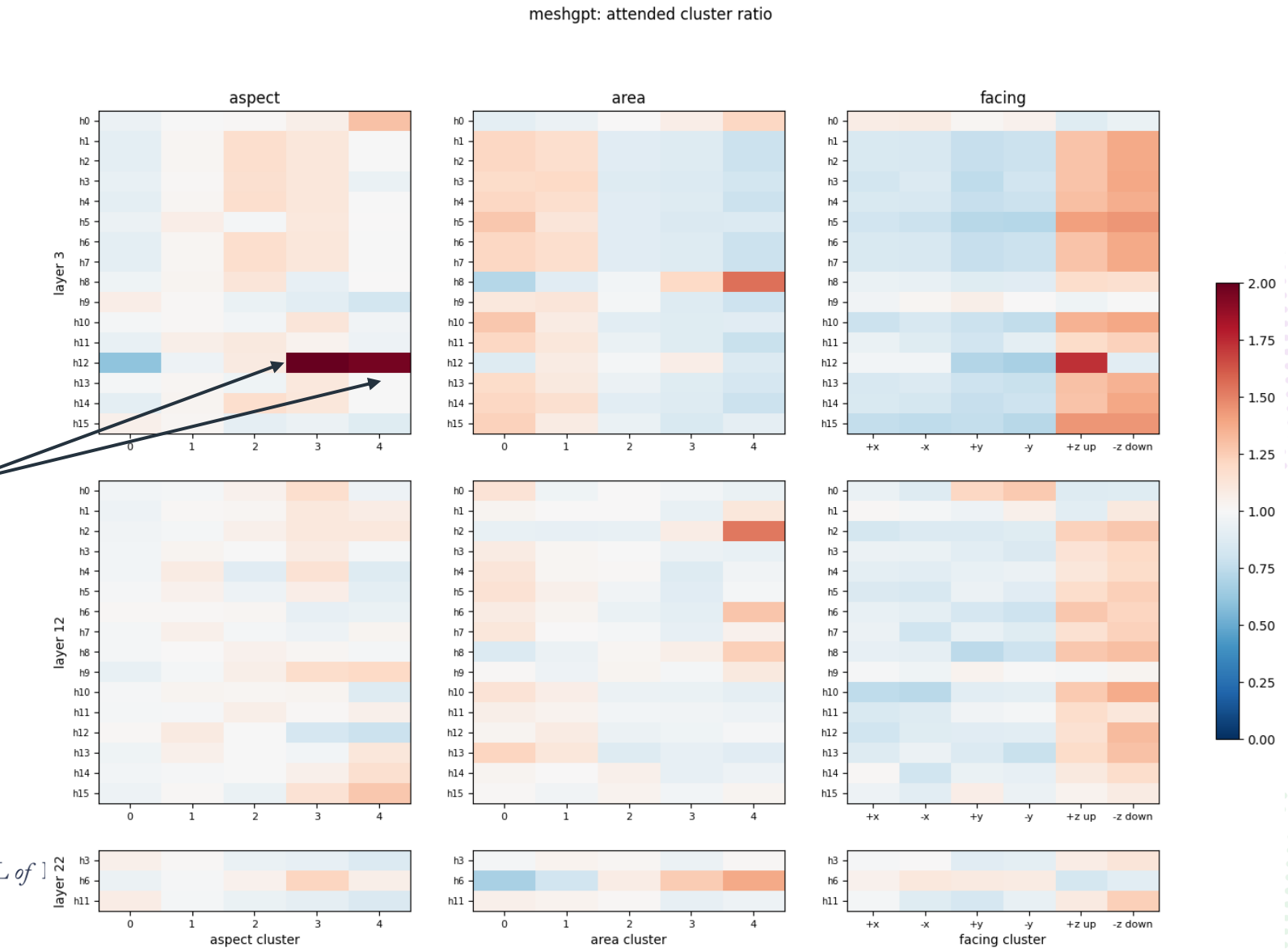
Attention Head Preferences

>1 means favor

1< means avoid

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MeshGPT Findings(WIP)



*Head 12 strongly favors **Thin & Elongated** Triangles at Layer 3*

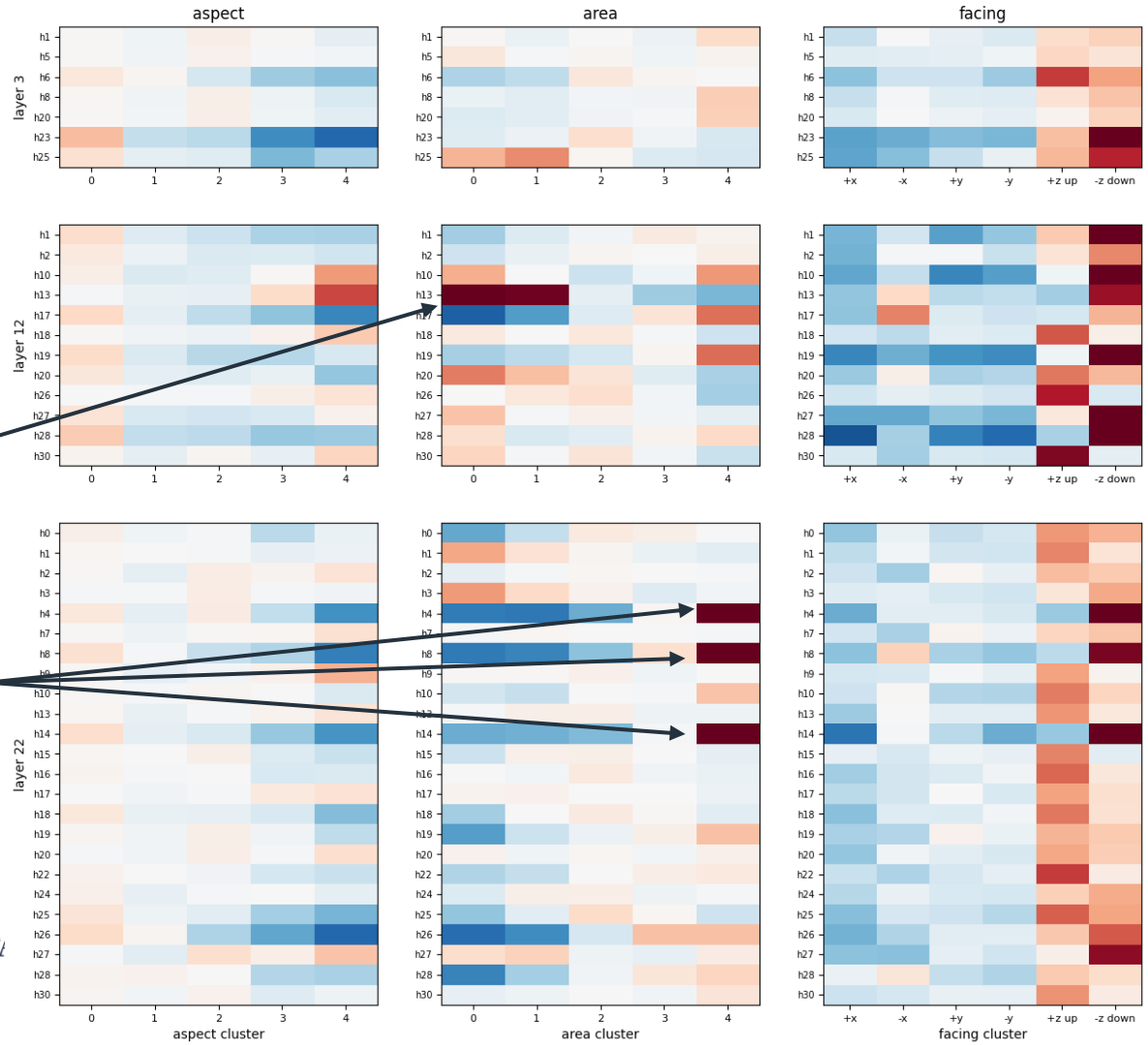
Attention Head Preferences

>1 means favor

1< means avoid

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MeshXL Findings (WIP)



Varying facing orientation tendencies

Attention Head Preferences

>1 means favor

1< means avoid

Head 13 ***Smaller Area*** *Triangles at Layer 12*

Head 4, 8, 14 ***Larger Area*** *Triangles at Layer 12*

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Extending the work

What's next after this investigation? (Current Ideas)

1. **Fine-tune 3D generators to generate artist quality mesh**
   - Amplify certain head behavior, injecting a prior/bias to certain heads to force them to produce consistent attention preference.
2. **Hard-coding programmable rules on attention heads**
   - For heads that demonstrate consistent preference, hard-code their behavior into rules/policies to reduce computational expensiveness or potentially leading to better generation quality.
3. **The why question**
   - Why does the model choose to attend to such features?

Challenges

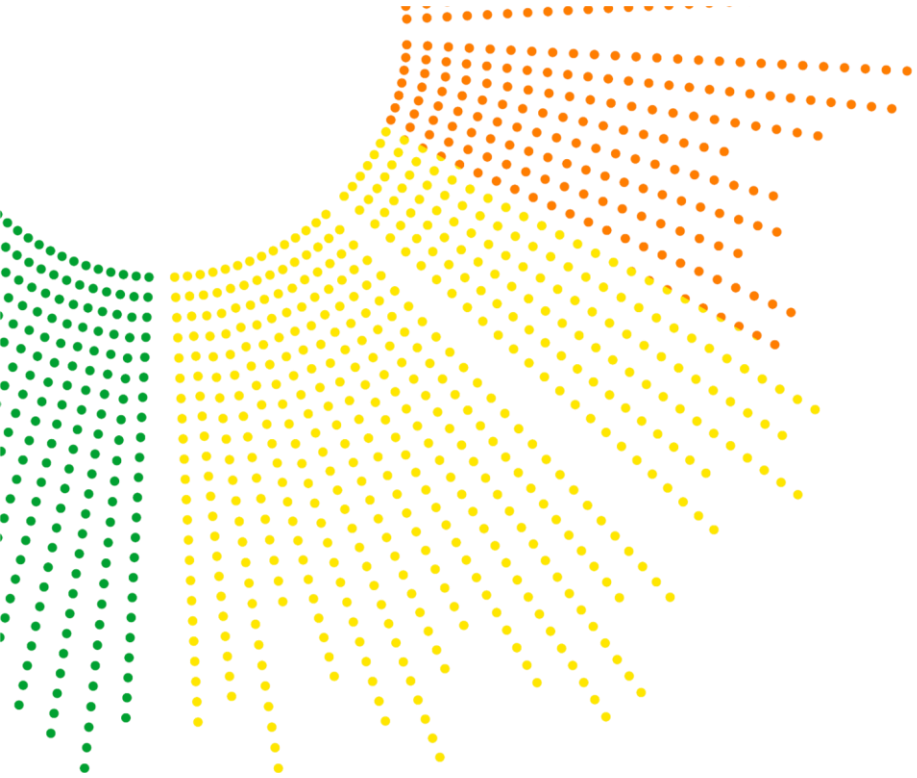
- Retraining/fine-tuning can be computationally expensive
- Uncertainty about which methods would be beneficial

Contact Info & background

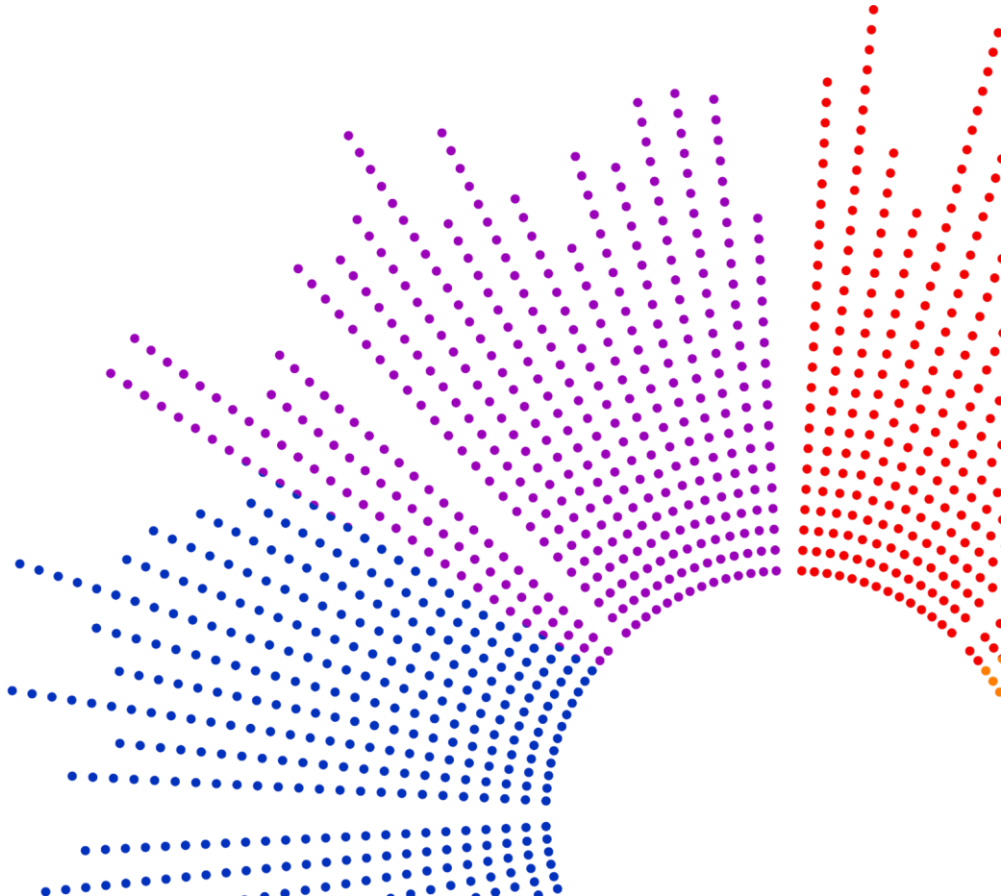
- 4th year UVA undergrad
- B.S. in Data Science Major (ML concentration)
- Pursuing Grad School Opportunities
- Email: wkt7ne@virginia.edu



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Thank you



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